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The effect of cognitive training in older adults: be aware of CRUNCH

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ABSTRACT

Cognitive training research has revealed support for both the magnification and compensation accounts, respectively predicting larger or smaller training benefits for individuals already having strong cognitive abilities. A recent study in older adults (OAs) revealed evidence for the Compensation-Related Utilization of Neural Circuits Hypothesis, which best fits a magnification account. Only few studies examined working memory (WM) training gains in OAs as a function of initial executive functioning level. Here, we tracked the progress of high and low functioning OAs across WM training sessions. The high functioning OAs continued to improve their performance on the three training tasks. However, for one of the training tasks, the low functioning participants failed to improve performance or terminated participation in the course of training. The results suggest that during cognitive training, evaluation of the individual's cognitive capacity and performance monitoring are essential to maintain an optimal balance between cognitive resources and task demands.

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Introduction

Aging-associated cognitive decline hampers the individual's wellbeing, quality of life, and successful participation in society. In the last decades, researchers have designed interventions to counteract this decline, especially working memory (WM) training. However, research on WM training of older adults (OAs), and on cognitive training in general, has evoked many contradictory findings and debates (Au et al., 2015; Borella, Carbone, Pastore, De Beni, & Caretti, 2017; Melby-Lervåg & Hulme, 2013; Melby-Lervåg, Redick, & Hulme, 2016; Morrison & Chein, 2011; Shipstead, Hicks, & Engle, 2012; Shipstead, Redick, & Engle, 2012). Most researchers agree that WM training can improve performance on the trained, or closely related, tasks but the controversy mainly concerns the claim that training also benefits transfer to non-trained cognitive tasks and domains. Some researchers are (moderately) positive about the existence of such transfer (Au et al., 2015; Karbach & Verhaegen, 2014; Morrison & Chein, 2011; Shah, Weinborn, Verdile, Sohrabi, & Martins, 2017), whereas others adopt a more critical perspective, for example, also criticizing previous studies and meta-analyses in terms of methodological flaws (Eikelboom,

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Bertens, & Kessles, *in press*; Green, Strobach, & Schubert, 2014; Melby-Lervåg & Hulme, 2013, 2016; Owen et al., 2010; Sprenger et al., 2013). Moreover, even in case of positive transfer effects, the question of the underlying mechanism remains. They may reflect enhanced WM ability or the enhancement of non-WM related, basic cognitive processes, such as overall speed or familiarity with the stimuli used, or the learning of some strategy (Melby-Lervåg et al., 2016; Minear et al., 2016; Morrison & Chein, 2011).

Traditional aging studies, including intervention studies, treat OAs as a homogenous group and conclusions are based on comparing old and young adults. However, given the remarkable variability in cognitive capacity at old ages (Christensen et al., 1999; Stern, 2009), it is important to take individual differences into account, especially when addressing the question who would benefit most from training. More recent research acknowledges the need for considering individual characteristics that might modulate training gains, such as age, educational attainment, motivation, personality, cognition-related beliefs, and baseline cognitive abilities (e.g., Borella et al., 2017; Guye, De Simoni, von Bastian, 2017; Jaeggi, Buschkuhl, Jonides, & Shah, 2011; Jaeggi, Buschkuhl, Shah, & Jonides, 2014; Von Bastian & Oberauer, 2014; Wiemers, Redick, & Morrison, 2019). The focus of the present study is on baseline cognitive abilities. Two accounts are relevant in this context. The magnification account states that the highest gains, either on the trained task, closely related tasks, and/or far-transfer tasks, are to be expected for individuals who already perform well. This account assumes that high performing individuals, with more cognitive resources, are better able to obtain and use new strategies and abilities. Instead, the compensation account assumes that low performing individuals will benefit most because of their larger potential for improvement (Lövdén, Brehmer, Li, & Lindenberger, 2012).

Previous training studies with OAs yielding data that speak to these two accounts have revealed mixed results, either supporting the compensation account (López-Higes et al., 2018; Roheger, Meyer, Kessler, & Kalbe, 2019; Witlock, McLaughlin, & Allaire, 2012; Zinke et al., 2014), the magnification account (Matysiak, Kroemeke, & Brzezicka, 2019), or both (depending on the type of transfer task: Borella et al., 2017). A recent study by Fu, Maes, Kessels, and Daselaar (2017) provided support for the magnification account. High-functioning OAs consistently profited from increasing difficulty levels of embedded aids in an ongoing task. However, OAs with relatively poor executive functions (EFs) showed an effect consistent with the Compensation-Related Utilization of Neural Circuits Hypothesis (CRUNCH), a hypothesis initially derived from neuroimaging studies (Cappell, Gmeindl, & Reuter-Lorenz, 2010; Reuter-Lorenz & Cappell, 2008; Schneider-Garces et al., 2010). The CRUNCH claims that OAs may reach a memory performance level that is comparable to that of younger adults by recruiting more prefrontal activation than younger adults. However, this compensatory mechanism is limited by the individual's cognitive capacity. Beyond a critical threshold, increasing cognitive demands results in a decreased activation of prefrontal brain areas and corresponding decreased performance. Accordingly, for the OAs with poor EF in Fu et al. (2017), stronger task requirements during encoding resulted in a subsequent decreased (memory) task performance, presumably because the task requirements went beyond their capacity. This suggests that cognitive training studies should consider the possibility of an effect as suggested by the CRUNCH. Especially when using adaptive training regimes that are commonly used in training studies, in which participants have to continuously adapt to increasing task demands, one could expect to see less, or even impaired, training and

transfer effects once the individual's maximum capacity has been reached. Moreover, according to the CRUNCH, OAs with relatively poor cognitive functions tend to reach this threshold earlier. Hence, this model calls for an evaluation of the individual's cognitive capacity before training and to constantly monitor performance on the training task(s).

The primary purpose of the present study was to test these predictions of the CRUNCH, focusing on the performance level across training sessions for OAs with different levels of baseline cognitive abilities. Specifically, as in Fu et al. (2017), we used two EF tasks to measure OAs' cognitive capacities and divided them into high and low EF groups. All participants underwent adaptive cognitive training sessions using three common computerized WM updating tasks. Each participant's training progress was continuously monitored, and training task difficulty was adjusted automatically based on the participant's performance. Training tasks' performances were tracked in terms of the score on the main performance measure of each task and in terms of dropout rate across training sessions. Based on Fu et al. (2017), we expected both low and high-functioning OAs to show a performance increase and continued participation in the first few training sessions. However, when the difficulty of the training tasks would surpass the cognitive capacity of the low-functioning OAs, we expected a performance decrease and/or dropout from the training. In contrast, this pattern was not expected to occur for high-functioning OAs. As a secondary aim of the present study, we also assessed training-induced transfer effects to non-trained memory tasks. For this purpose, the participants performed two ecologically relevant memory tasks before and immediately after training. In theory, the expected EF-dependent differential training gains could also induce differential transfer benefits. Specifically, the stronger training-task performance increment expected for the high-functioning OAs relative to the low-functioning OAs could also imply larger transfer effects for the former compared to the latter OAs, which would provide further support for the CRUNCH model. However, based on the null results of many previous studies assessing the transfer of cognitive trainings to non-trained tasks, one could expect limited transfer effects overall.

Methods

Participants

Fifty-one OAs, recruited by an advertisement in local newspapers, participated in the study. Three participants dropped out before the start of the experiment due to personal reasons. Participants were divided into a low and high EF group using a median split based on the composite z-score of neuropsychological tests (Fu et al., 2017; see below). The participants from each group were invited to complete 20 training sessions (see below). Due to missing data (i.e., due to not having performed sufficient training sessions), we ended up with 42 participants (20 low EF and 22 high EF) of which we had sufficient data (i.e., of at least 12 sessions) for at least one of the three tasks used for training. One participant from the low EF group did not perform any 2-back and numerical updating task sessions. Two other low EF participants did not complete the post-training assessment of one of the transfer tasks (see below). All participants signed an informed consent form and received €50 as remuneration. The study was approved by the Ethics Committee of the Faculty of Social Sciences of Radboud University.

Materials

Pre-training screening

Mini-Mental State Examination (MMSE). The MMSE (Folstein, Folstein, & McHugh, 1975) was used as screening instrument to obtain a global impression of the cognitive functioning of the participants. The instrument consists of 20 items and the score ranges from 0 – 30 points, with 25 – 30 points reflecting normal functioning. This screening was only performed during the pre-training assessment.

Pre- and post-training assessments

The following tests were applied both before and after training, to assess transfer effects to non-trained/practiced tasks.

EF capacity tests. As in Glisky, Polster, and Routhieaux (1995), three validated and widely used neuropsychological tests were used to measure EF. These tests were used both to divide the participants into a high- and low-functioning group and to assess training-induced transfer effects. The following tests were employed: 1) Backward Digit Span from the Wechsler Adult Intelligence Scale – Fourth Edition (WAIS-IV; Wechsler, 2008), 2) Mental Arithmetic, and 3) Mental Control from the Wechsler Memory Scale – Fourth Edition (WMS-IV; Wechsler, 2009). However, we did not use the participants' performance on the Mental Control test for determining the composite EF score due to a ceiling effect (Full score = 8, Median = 8, Mean = 7.58, $SD = 1.23$). Pearson correlation between the performance measures of the two other tests was $r = .37$, $p = .02$. Performance on these two tests was converted to a z score. Subsequently, the average of the two z scores was taken as a composite EF score, representing each individual's EF capacity.

Faces and names task. Twelve faces, each coupled with a Dutch surname, were presented to the participants. Each face was displayed on a computer screen for 30 s, after which the next face and name were immediately presented. Subsequently, the faces were again displayed, but this time without the names. The participants were asked to recall the correct face-name combinations. Two different lists were used in the pre- and post-training tests. The outcome measure of this task was the number of correctly recalled face-name combinations.

Shopping list task. Participants were handed a shopping list containing 20 items. These 20 items could be divided into five categories, for example, fruits, tools, and clothes. The items were placed randomly on the shopping list and each participant was allowed to read the list for 5 min. Immediately thereafter, the participant was given 5 min to write down all the items he or she could recall. Two different lists were used in the pre- and post-training tests. The performance measure was the number of items correctly recalled.

Training tasks

Like in the COGITO study (Schmiedek, Lövdén, & Lindenberger, 2010), three WM updating tasks were used for training. All tasks were programmed in PsychoPy.

Alpha-span task. Upper-case consonants were sequentially presented, with a number presented below each letter. For each letter, participants had to decide as quickly as possible whether the number below corresponding to the position in the alphabet of the current letter within the set of letters presented up to this step. The participant had to press the letter “V” in case the number corresponded with the alphabetic position (half of the trials) and the letter “B” in case the number differed from the correct position. The position number deviated by ± 1 when the number did not match the alphabetic position. Based on the performance in the previous block, the sequence length for each block was individually adjusted. The first sequence was the same for all participants and had a length of two letters. If the participant answered all of the trials correctly, one more letter was added to the next sequence. The next sequence contained one letter less in case of $>55\%$ incorrect answers. In each daily session, participants were asked to finish one block of 15 trials. The participants needed 4 – 8 min per day to perform the task.

2-back task. In a sequence of 20 trials, a picture of the puppet Mario appeared at one of four possible locations on the screen. It was the participant’s task to recognize whether or not each picture appeared in the same position as it was two trials earlier in the sequence (2-back). The participant had to press “V” and “B” in case of, respectively, a matching and non-matching position. The pictures appeared at random locations for each trial. Participants completed three sequences each day. Based on the performance on the previous sequence, the presentation rate of the pictures during the current sequence was individually adjusted by varying the inter-stimulus interval (ISI): the interval between onset of the stimulus and onset of the next stimulus, which is independent of the participant’s response. Every participant started with an inter-stimulus interval (ISI) of 5000 ms. When the participant gave a correct response on $>90\%$ of the trials, the ISI for the next sequence was reduced by 200 ms. In case of $<60\%$ correct responses, the ISI was increased by 200 ms in the next sequence. Participants took about 3 – 5 min per day for this task.

Numerical updating task. On each trial, three single numbers, ranging from 0 to 9, were simultaneously and horizontally presented in the upper three cells of a 2×3 grid. The numbers were presented for 5 s, after which they disappeared. The participant was instructed to remember the numbers. Subsequently, a number within a range of -8 to $+8$ was presented in the cell underneath one of the numbers. The participant was instructed to add or subtract the value of this number to/from the value of the number that had been presented in the cell right above. The participant had to memorize the updated value for this cell. All intermediate and end results ranged between 0 and 9. Each sequence contained 2 – 8 updating operations, using an ISI of 5000 ms. After each such sequence, the participant was asked to enter the value of each of the three updated numbers (top three cells), using the keyboard. The participant had to complete 15 sequences on each daily session, separated in three blocks. Based on the performance of a previous block of sequences, the presentation time for each updating operation was individually adjusted. All participants started with an ISI of 5000 ms. If participants consistently could give the correct answer in five sequences (one block), the ISI was reduced by 200 ms. The presentation time was increased with 200 ms in case of three mistakes in a row. It took the participant 3 – 8 min to complete each daily task.

Procedure

The pre-training assessment, consisting of the MMSE, EF, and memory tests, were first conducted in a randomized order for each subject. Subsequently, the training program was installed on the participant's PC. The participant was then instructed on how to run the training program and could practise. Each participant received a printed list that stated the order of the training tasks that he or she should follow in each training session. The participants were asked to finish the 20 training sessions within 30 days after the pre-training assessments. Training progress of each participant was monitored by Dropbox shared folder. Participants would receive a reminder by the researcher if no login file of the training program was generated 3 days in a row. The participants performed the EF and memory tests once again after the training.

Data analysis

All statistical analyses were performed using SPSS23. Differences between the high and low EF groups on demographic and pre-training assessment variables were tested using analysis of variance (ANOVA). The (differential) change in performance between pre- and post-training performance for the Backward Digit Span, Mental Arithmetic, Faces & Names, and Shopping List tasks was evaluated using Group (high vs. low EF) $\times$ Test (pre- vs. post-training test) repeated measures (RM) ANOVAs. Training progress on each training task was analyzed in two ways. First, 23 participants terminated their participation in one (17 participants), two (3 participants), or all three (3 participants) of the training tasks at some point during the training.

To assess group differences in the dropout rate, the significance of the between-group difference in mean number of completed sessions was tested with a Mann–Whitney U test. Second, given the relatively many missing data points, we applied linear mixed models (LMM) to analyze progress in task performance across the 20 training sessions. We used the highest score obtained within a session as performance index. Thus, the index reflected the longest span for the alpha-span task, and the shortest inter-stimulus interval (ISI) for the 2-back and numerical updating tasks in each session. For each task, we started off with a model with Group (as factor), Session (as covariate), Session $\times$ Session (also as covariate, to capture possible quadratic trends), and the respective interactions as both fixed and random effects, including the intercepts. The correlation in task performance between sessions revealed a pattern of decreasing correlations with increasing distance between two sessions. Therefore, for the repeated Session and Session $\times$ Session effects we assumed a first-order autoregressive (AR[1]) covariance structure matrix. For the random effects, we chose the variance component procedure. The restricted maximum likelihood (RML) estimation approach was used in all analyses. By systematically eliminating fixed and/or random effects we determined the model with the best fit, using the BIC (Bayesian information criterion). In building and evaluating the various models and in case of no large differences in model fit, we prioritized interaction effects, given our hypotheses. All statistical significance tests were 2-tailed with $\alpha = 0.05$. Reported confidence intervals (CI) are 95% CI.

Results

Descriptive statistics and pre-post training analyses

Table 1 summarizes the demographic details and pre- and post-training test results. Regarding the pre-training assessment, the low and high EF groups significantly differed on performance on the Backward Digit Span and Mental Arithmetic subtests, but not on any of the other variables. The RM-ANOVAs did not reveal any significant effects for Session or the Group $\times$ Session interaction.

Training progress

Alpha-span task

The top right panel of Figure 1 shows, for each training session, the percentage of participants from each group that completed the alpha-span task. One participant from the high EF group failed to complete the last training session. In the low EF group, two participants dropped out from Session 13 onwards, another participant stopped participating in this task from Session 14, etc. Sixty-five percent of the low EF OAs completed all 20 sessions. The difference in the mean number of completed training sessions between the high EF group ($M = 19.96$, $SD = 0.21$) and low EF group ($M = 18.40$, $SD = 2.87$) was significant Mann–Whitney $U = 150$, $p = .01$, reflecting the larger dropout in the low EF group. The top left panel shows the progress in task performance across training sessions. The means are based on all participants that performed the corresponding session. The

Table 1. Demographics and pre- and post-training test results for the two groups.

	Low EF	High EF	F^c	p (2-tailed)
N	20	22		
Gender (female)	10	11		
Age	67.90 (5.25)	67.95 (5.92)	0.00	0.98
Education^a				
Level 4	4	1		
Level 5	8	7		
Level 6	7	8		
Level 7	1	6		
MMSE^b	29.30 (1.49)	29.86 (0.47)	2.85	0.10
Backward Digit Span				
Pre-training	7.35 (1.14)	9.95 (1.05)	59.84	< 0.001
Post-training	8.40 (2.58)	10.18 (1.44)	3.82/1.59	0.06/0.22
Mental Arithmetic				
Pre-training	14.05 (1.76)	16.64 (2.38)	15.74	< .001
Post-training	13.85 (3.92)	17.23 (2.56)	0.28/1.16	0.60/0.29
Faces & Names				
Pre-training	2.78 (2.18)	3.23 (2.53)	0.35	0.56
Post-training	2.72 (1.57)	4.41 (1.76)	3.09/3.74	0.09/0.06
Shopping List				
Pre-training	14.35 (4.45)	16.00 (3.68)	1.73	0.20
Post-training	13.65 (4.77)	15.00 (4.36)	2.83/0.09	0.10/0.77

^aEducation level is based on the Dutch educational system using a 7-point scale (1 = less than primary school; 2 = primary school; 3 = secondary education with no diploma; 4 = lower secondary vocational education; 5 = intermediate vocational education; higher vocational/professional education; 7 = academic degree; Verhage, 1964). ^bMMSE: Mini-Mental State Examination. ^c F : the values under post-training represent F values for the Session and Group $\times$ Session factors, respectively, with associated p -values in the respective column. The degrees of freedom (df) for the denominator of the F statistic are 40 in all analyses, except for the pre-post training analysis of the Faces & Names task, for which $df = 38$. Values in parentheses are standard deviations.

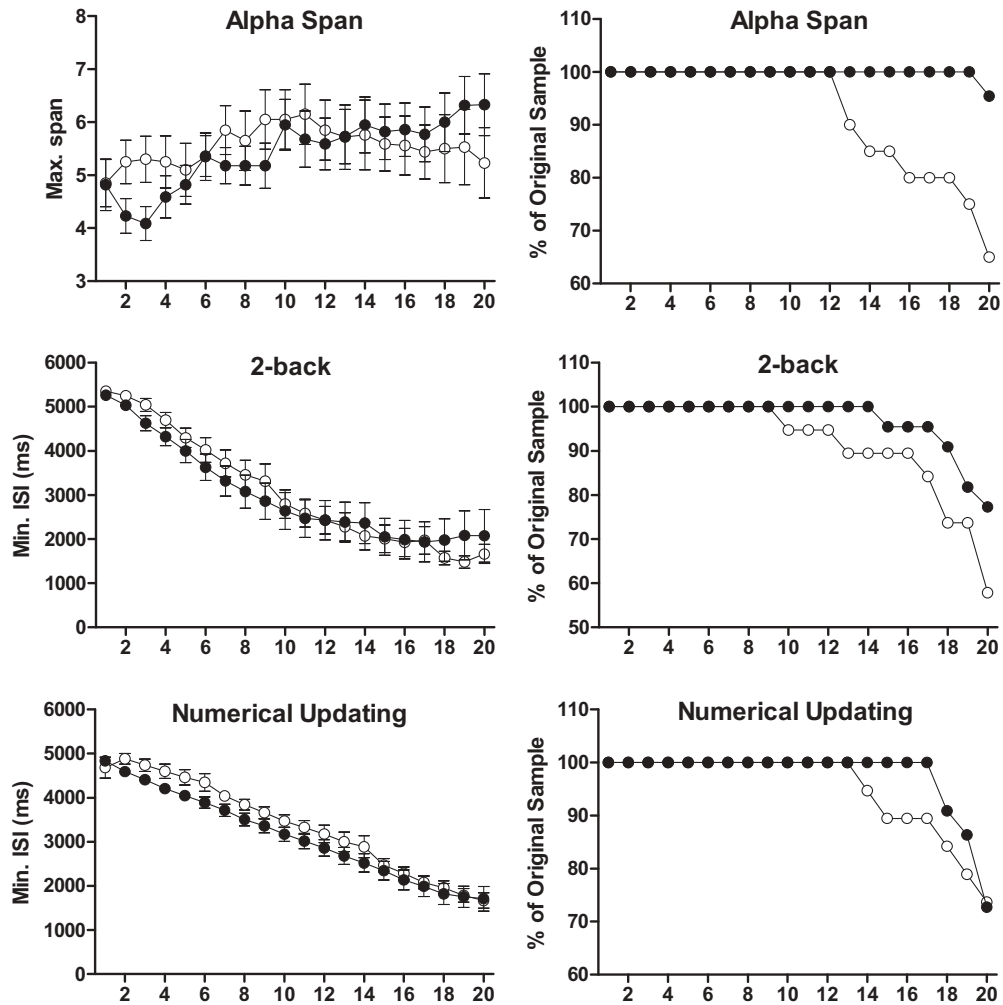


Figure 1. Top left: Group's mean performance ($\pm$ standard error of the mean; SEM) during each session of the alpha-span task. Top right: percentage of original group size of individuals participating in each session. Middle left and right, and bottom left and right panels: same as corresponding top panels but for, respectively, the 2-back and numerical updating task. Filled and open circles represent, respectively, high EF participants and low EF participants.

figure suggests a gradual performance increase for the high EF but not low EF OAs. LMM analyses revealed a best-fitting model with the Group $\times$ Session interaction as fixed and random effects and including the intercepts (7 model parameters). The fixed Group $\times$ Session interaction effect was significant in this model, $F(2, 42.76) = 5.55$, $p = .007$, reflecting a significant Session effect for the high EF participants, $t(40.78) = 3.03$, $p = .004$, parameter estimate $E = .08$, $SE = .03$, $CI[0.03, 0.14]$, but not low EF OAs, $E = .05$, $SE = .03$, $CI[-0.02, 0.11]$, $p = .14$. As can be seen in the graph, the high EF OAs showed a performance decrease in Sessions 2 and 3. One possible implication of this is that the significant effect of Session for this group, but not the low EF group, simply reflects a catching up of the high EF OAs to the performance level of the low EF participants. To

investigate this, we performed an identical LMM analysis as reported above but now using Session 4, on which the two groups again displayed a non-significant performance difference, as starting point. The LMM analysis on Sessions 4 – 20 revealed the same results as that performed on Sessions 1 – 20: a session effect, reflecting increasing performance across training sessions, for the high EF but not low EF group. Hence, the high EF group's performance increase went above simply approaching the low EF group's level.

2-back task

The middle right panel shows the dropout pattern across the 2-back task sessions. The percentage of participants that completed all training sessions was 57.9 for the low EF OAs and 77.3 for the high EF participants. The number of completed 2-back task sessions ($M = 19.36$, $SD = 1.47$, for the high EF, and $M = 18.32$, $SD = 3.06$ for the low EF group) did not significantly differ between the two groups, Mann–Whitney $U = 166$, $p = .17$. As depicted in the middle left panel, both groups displayed a steady decrease in ISI until about the middle of the training, followed by a slowed-down decrease toward the end of the training. This pattern is also reflected in the outcome of the LMM analyses. These revealed a best-fitting model with both Session and Session $\times$ Session (quadratic trend) as fixed and random effects, including the corresponding intercepts (7 model parameters). In this model, the effect of both Session and Session $\times$ Session were significant $F_s > 160.02$, $p_s < .011$. The corresponding parameter estimates were $E = -.38$, $SE = .03$, $CI[-0.43, -0.32]$, $t(94.08) = -13.66$, $p < .001$ for Session, and $E = .01$, $SE = .00$, $CI[0.009, 0.012]$, $t(606,38) = 12.65$, $p < .001$, for Session $\times$ Session.

Numerical updating task

The bottom right panel displays the dropout rates across sessions for the numerical updating task. The figure suggests no large group difference in mean number of completed sessions ($M = 19.50$, $SD = 0.96$, for the high EF, and $M = 19.00$, $SD = 2.11$ for the low EF OAs), which was statistically supported by a non-significant group effect, Mann–Whitney $U = 204$, $p = .87$. As can be seen in the figure in the bottom left panel, both groups showed a steady decrease of the minimal achieved ISI across training sessions. LMM analyses revealed a best-fitting model that contained Session as fixed and random effect, with only the fixed effect intercept (5 model parameters). The fixed effect of Session was significant, $F(1, 52.15) = 274.88$, $p < .001$, with a corresponding parameter estimate of $E = -.16$, $SE = .01$, $CI[-.18, -.14]$, $t(52.15) = -16.58$, $p < .001$.

Discussion

The present study examined a possible modulatory role of individual differences in cognitive capacity on potential benefits of a training program, targeting WM updating ability, on performing the trained tasks and two ecologically valid transfer tasks. A sample of OAs was split into a high and low EF group. Both groups continuously improved on the 2-back and numerical updating tasks during training. However, across the alpha-span task sessions, the low EF OAs displayed a larger dropout rate than the high EF OAs. Moreover, the latter OAs showed a significant performance improvement across the training sessions, whereas the former OAs did not. However, we failed to find any (differential) benefits of the training with respect to performance on the non-trained, transfer tasks.

One of the major questions in the cognitive training field relates to the selection of those individuals who would benefit most from cognitive training. The magnification account states that “the richer will get richer”, whereas the compensation account states that low-functioning individuals have more room for improvement (“the poor will get richer”). Our results are consistent with the magnification account. The high EF group showed a consistent improvement in each of the three training tasks. Given the lack of a follow-up assessment, we have no information on the maintenance of these training gains, which is a limitation of the present study. But most importantly, like in Fu et al. (2017), we observed a pattern that we would like to argue to be consistent with the CRUNCH in the low EF OAs during training of the alpha-span task. Like the high EF OAs, these participants showed a consistent increase in performance during the first 12 training sessions, when none of the OAs had terminated participation in this task yet. In fact, a RM-ANOVA using Group as between-subjects factor and Session (12) as repeated measure on the performance index of this task only revealed a highly significant Session effect ($p < .001$, other $ps > .35$). However, from training Session 12 on, some of the low EF participants started to drop out, while the remaining low EF participants did not show any further training improvement. Presumably, by Session 12 the difficulty of the alpha-span task had increased beyond this group’s capacity, resulting in a decreasing ability and motivation to perform the task. Notably, the dropouts occurred despite monitoring and sending reminders. Without such monitoring, the dropout rate would likely have been even higher (e.g., Vermeij, Claassen, Dautzenberg, & Kessels, 2016).

The pattern of results for the low EF OAs can also be evaluated in the framework of the concept of supply-demand mismatch proposed by Lövdén, Bäckman, Lindenberger, Schaefer, and Schmiedek (2010). This concept entails that training-induced changes are a function of the difference between available cognitive resources and task demands: changes occur only in case demands exceed resources. However, the present results suggest a change in the supply-demand difference across the training in the low EF group. Accordingly, in the first 12 sessions, the difference was such that the participants were willing and able to increase their cognitive effort to adapt to the (adaptive) training protocol. However, on Session 12, demands had reached a critical point at which the participants were no longer able or willing to invest any further effort. This performance pattern suggests that the supply-demand mismatch can also become too large, preventing further increases or even prompting decreases in the use of cognitive resources. Instead, for high EF OAs, the supply-demand mismatch remained constant and optimal across the training. Notably, the two groups showed similar training benefits across training on each of the other two training tasks. Presumably, the implied supply-demand mismatch was the same and relatively constant (with demands exceeding resources) for these tasks in each of the two groups, resulting in a steady task performance improvement. In other words, for both the low and high EF OAs, the task difficulty remained not too far removed from the available cognitive resources to support training effects. In this context, it is worthy of note that in these two tasks we used the ISI to manipulate task difficulty. This might imply that these tasks trained (more simple) response speed rather than (more complex) WM, which might be the reason why we did not observe a CRUNCH performance pattern in these tasks. This is a limitation of the present study and future studies should assess whether, for example, adapting the n-back level rather than ISI would reveal a CRUNCH pattern for at least low EF OAs. However, 3-

and 4-back tasks are often too demanding, even for OAs with assumed strong cognitive capacity, which may cause floor performances.

These considerations imply that the mixed results regarding the validity of the magnification versus compensation account could at least partly be due to between-study differences in the magnitude of the difference between cognitive resources and task difficulty. Accordingly, if that difference is within reasonable limits and consistently remains so across training, those with lower cognitive capacities (having more room for improvement) might show larger training improvements than those with stronger capacities, supporting the compensation account. However, when the demand exceeds a certain limit, implicating a demand-resource difference that is too large, training benefits are limited or absent, and this point is reached sooner in OAs with relatively weak cognitive resources, supporting the magnification account. Notably, the CRUNCH model that we put forward here as supporting the magnification account is also not entirely inconsistent with the compensation account. If provided with training tasks that are within the limits of the participant's cognitive resources, for example, a task that maintains an adequate (relatively low) difficulty level, lower functioning OAs would potentially benefit from this and reach the same performance level as high-functioning OAs. High-functioning OAs, in turn, would not show a beneficial effect, because they already would perform at an optimal level.

We failed to find any evidence of transfer of training benefits to the tasks used for the original screening and to two non-trained (episodic) working memory tasks. These null results suggest that the differences in training benefits between the low and high EF OAs seen for one of the tasks did not result in differential training-induced transfer benefits and add to the lack of, or very limited, transfer effects seen in the process-based cognitive training literature in general (e.g., Melby-Lervåg et al., 2016). We used a daily training duration of 10 – 21 min, implying a mean total training time of about 5 h in our study. The main reason to choose this duration was to optimize motivation and adherence. The feasibility of a study with longer daily training times is likely to be low, with poor adherence and an even higher expected dropout than in the present study. One could argue that longer training sessions might have resulted in better training gains, but it is difficult to a-priori determine the optimal training duration. The duration of WM training in previous studies in OAs varies considerably and the results of meta-analyses are inconsistent (see review by Teixeira-Santos et al., 2019, reporting a range of 1.5–20 h of total training time, and 3 – 40 training sessions). For example, these studies also reveal some evidence of *smaller* gains when dose (in hours) and training length (in weeks) increase (e.g., see Table 2 in Teixeira-Santos et al., 2019). Next, to the issue of training time, future research examining training-induced transfer effects for OAs with different pre-training cognitive abilities should include an appropriate control group to be able to estimate the magnitude of such effects and potential differences therein between high- and low-functioning OAs. Even in case such future research would continue to fail to provide evidence of (differential) transfer benefits, one could still argue that a consideration of the CRUNCH is relevant in the context of the trained task itself. Specifically, the present results suggest that in training some important (new or lost) daily-life skill, the aimed-for achievement level should not excessively exceed the individual's cognitive capacity.

Conclusions

Based on our previous study (Fu et al., 2017), we tested predictions derived from the CRUNCH model in the framework of the question which OAs would show the largest improvements during a WM training program: those with relatively strong or poor EFs. As hypothesized from the model, low functioning but not high-functioning OAs showed a CRUNCH pattern during training of the alpha-span task. The results are in line with the magnification account, according to which high-functioning OAs should show the strongest training benefits. An implication of these findings is that, in providing cognitive training programs, it is important to match the difficulty of the training tasks to the cognitive ability of the trained individual. Especially when conducting training studies on elderly participants, a close performance monitoring is essential to assess the point at which the individual reaches the CRUNCH threshold and a further increase in task difficulty is counterproductive.

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References

- Au, J., Sheehan, E., Tsai, N., Duncan, G. J., Buschkuhl, M., & Jaeggi, S. M. (2015). Improving fluid intelligence with training on working memory: A meta-analysis. *Psychonomic Bulletin and Review*, 22, 366–377. doi:10.3758/s13423-014-0699-x
- Borella, E., Carbone, E., Pastore, M., De Beni, R., & Caretti, B. (2017). Working memory training for healthy older adults: The role of individual characteristics in explaining short- and long-term gains. *Frontiers in Human Neuroscience*, 11, 99. doi:10.3389/fnhum.2017.00099
- Cappell, K. A., Gmeindl, L., & Reuter-Lorenz, P. A. (2010). Age differences in prefrontal recruitment during verbal working memory maintenance depend on memory load. *Cortex*, 46, 462–473. doi:10.1016/j.cortex.2009.11.009
- Christensen, H., Mackinnon, A. J., Korten, A. E., Jorm, A. F., Henderson, A. S., Jacomb, P., & Rodgers, B. (1999). An analysis of diversity in the cognitive performance of elderly community dwellers: Individual differences in change scores as a function of age. *Psychology and Aging*, 14, 365–379. doi:10.1037/0882-7974.14.3.365

- Eikelboom, W. S., Bertens, D., & Kessles, R. P. C. (in press). Cognitive rehabilitation in normal aging and individuals with subjective cognitive decline. In N. Chiaravalloti, J. DeLuca, & E. Weber (Eds.), *Cognitive rehabilitation and neuroimaging: Examining the evidence from brain to behavior*. New York: Springer.
- Folstein, M. F., Folstein, S. E., & McHugh, P. R. (1975). Mini-mental state. *Journal of Psychiatric Research*, 12, 189–198. doi:10.1016/0022-3956(75)90026-6
- Fu, L., Maes, J. H. R., Kessels, R. P. C., & Daselaar, S. M. (2017). To boost or to CRUNCH? Effect of effortful encoding on episodic memory in older adults is dependent on executive functioning. *PLoS One*, 12, e0174217. doi:10.1371/journal.pone.0174217
- Glisky, E. L., Polster, M. R., & Routhieaux, B. C. (1995). Double dissociation between item and source memory. *Neuropsychology*, 9, 229–235. doi:10.1037/0894-4105.9.2.229
- Green, C. S., Strobach, T., & Schubert, T. (2014). On methodological standards in training and transfer experiments. *Psychological Research*, 78, 756–772. doi:10.1007/s00426-013-0535-3
- Guye, S., De Simoni, C., & von Bastian, C. C. (2017). Do individual differences predict change in cognitive training performance? A latent growth curve modeling approach. *Journal of Cognitive Enhancement*, 1, 374–393. doi:10.1007/s41465-017-0049-9
- Jaeggi, S. M., Buschkuhl, M., Jonides, J., & Shah, P. (2011). Short- and long-term benefits of cognitive training. *Proceedings of the National Academy of Sciences*, 108, 10081–10086. doi:10.1073/pnas.1103228108
- Jaeggi, S. M., Buschkuhl, M., Shah, P., & Jonides, J. (2014). The role of individual differences in cognitive training and transfer. *Memory and Cognition*, 42, 464–480. doi:10.3758/s13421-013-0364-z
- Karbach, J., & Verhaegen, P. (2014). Making working memory work: A meta-analysis of executive-control and working memory training in older adults. *Psychological Science*, 25, 2027–2037. doi:10.1177/0956797614548725
- López-Higes, R., Martín-Aragoneses, M. T., Rubio-Valdehita, S., Delgado-Losada, M. L., Montejo, P., Montenegro, M., ... López-Sanz, D. (2018). Efficacy of cognitive training in older adults with and without subjective cognitive decline is associated with inhibition efficiency and working memory span, not with cognitive reserve. *Frontiers in Aging Neuroscience*, 10, 23. doi:10.3389/fnagi.2018.00023
- Lövdén, M., Bäckman, L., Lindenberger, U., Schaefer, S., & Schmiedek, F. (2010). A theoretical framework for the study of adult cognitive plasticity. *Psychological Bulletin*, 136, 659–676. doi:10.1037/a0020080
- Lövdén, M., Brehmer, Y., Li, S. C., & Lindenberger, U. (2012). Training-induced compensation versus magnification of individual differences in memory performance. *Frontiers in Human Neuroscience*, 6, 1–14. doi:10.3389/fnhum.2012.00141
- Matysiak, O., Kroemeke, A., & Brzezicka, A. (2019). Working memory capacity as a predictor of cognitive training efficacy in the elderly population. *Frontiers in Aging Neuroscience*, 11, 126. doi:10.3389/fnagi.2019.00126
- Melby-Lervåg, M., & Hulme, C. (2013). Is working memory training effective? A meta-analytic review. *Developmental Psychology*, 49, 270–291. doi:10.1037/a0028228
- Melby-Lervåg, M., & Hulme, C. (2016). There is no convincing evidence that working memory training is effective: A reply to Au et al. (2014) and Karbach and Verhaeghen (2014). *Psychonomic Bulletin and Review*, 23, 324–330. doi:10.3758/s13423-015-0862-z
- Melby-Lervåg, M., Redick, T. S., & Hulme, C. (2016). Working memory training does not improve performance on measures of intelligence or other measures of “far transfer”: Evidence from a meta-analytic review. *Perspectives on Psychological Science*, 11, 512–534. doi:10.1177/1745691616635612
- Minear, M., Brasher, F., Brandt Guerrero, C., Brasher, M., Moore, A., & Sukeena, J. (2016). A simultaneous examination of two forms of working memory training: Evidence for near transfer only. *Memory & Cognition*, 44, 1014–1037. doi:10.3758/s13421-016-0616-9
- Morrison, A. B., & Chein, J. M. (2011). Does working memory training work? The promise and challenges of enhancing cognition by training working memory. *Psychonomic Bulletin & Review*, 18, 46–60. doi:10.3758/s13423-010-0034-0

- Owen, A. M., Hampshire, A., Grahn, J. A., Stenton, R., Dajani, S., Burns, R. J., ... Ballard, C. G. (2010). Putting brain training to the test. *Nature*, 465, 775–778. doi:10.1038/nature09042
- Reuter-Lorenz, P. A., & Cappell, K. A. (2008). Neurocognitive aging and the compensation hypothesis. *Current Directions in Psychological Science*, 17, 177–182. doi:10.1111/j.1467-8721.2008.00570.x
- Roheger, M., Meyer, J., Kessler, J., & Kalbe, E. (2019). Predicting short- and long-term cognitive training success in healthy older adults: Who benefits? *Aging, Neuropsychology, and Cognition*, 1–19. Advance Online Publication. doi:10.1080/13825585.2019.1617396
- Schmiedek, F., Lövdén, M., & Lindenberger, U. (2010). Hundred days of cognitive training enhance broad cognitive abilities in adulthood: Findings from the COGITO study. *Frontiers in Aging Neuroscience*, 2, 27. doi:10.3389/fnagi.2010.00027
- Schneider-Garcés, N. J., Gordon, B. A., Brumback-Peltz, C. R., Shin, E., Lee, Y., Sutton, B. P., ... Fabiani, M. (2010). Span, CRUNCH, and beyond: Working memory capacity and the aging brain. *Journal of Cognitive Neuroscience*, 22, 655–669. doi:10.1162/jocn.2009.21230
- Shah, T. M., Weinborn, M., Verdile, G., Sohrabi, H. R., & Martins, R. N. (2017). Enhancing cognitive functioning in healthy older adults: A systematic review of the clinical significance of commercially available computerized cognitive training in preventing cognitive decline. *Neuropsychology Review*, 27, 62–80. doi:10.1007/s11065-016-9338-9
- Shipstead, Z., Hicks, K. L., & Engle, R. W. (2012). Working memory training remains a work in progress. *Journal of Applied Research in Memory and Cognition*, 1, 217–219. doi:10.1016/j.jarmac.2012.07.009
- Shipstead, Z., Redick, T. S., & Engle, R. W. (2012). Is working memory training effective? *Psychological Bulletin*, 138, 628–654. doi:10.1037/a0027473
- Sprenger, A. M., Atkins, S. M., Bolger, D. J., Harbison, J. I., Novick, J. M., Chrabaszcz, J. S., ... Dougherty, M. R. (2013). Training working memory: Limits to transfer. *Intelligence*, 41, 638–663. doi:10.1016/j.intell.2013.07.013
- Stern, Y. (2009). Cognitive reserve. *Neuropsychologia*, 47, 2015–2028. doi:10.1016/j.neuropsychologia.2009.03.004
- Teixeira-Santos, A. C., Moreira, C. S., Magalhães, R., Magalhães, C., Pereira, D. R., Leite, J., ... Sampaio, A. (2019). Reviewing working memory training gains in healthy older adults: A meta-analytic review of transfer for cognitive outcomes. *Neuroscience and Biobehavioral Reviews*, 103, 163–177. doi:10.1016/j.neubiorev.2019.05.009
- Verhage, F. (1964). *Intelligentie en leeftijd onderzoek bij Nederlanders van twaalf tot zevenenzeventig jaar* [Intelligence and age: Research study in Dutch individuals age twelve to seventy-seven]. Assen: Van Gorcum/Prakke & Prakke.
- Vermelij, A., Claassen, J. A. H. R., Dautzenberg, P. L. J., & Kessels, R. P. C. (2016). Transfer and maintenance effects of online working-memory training in normal aging and mild cognitive impairment. *Neuropsychological Rehabilitation*, 26, 783–809. doi:10.1080/09602011.2015.1048694
- Von Bastian, C. C., & Oberauer, K. (2014). Effects and mechanisms of working memory training: A review. *Psychological Research*, 78, 803–820. doi:10.1007/s00426-013-0524-6
- Wechsler, D. (2008). *Wechsler adult intelligence scale – Fourth edition: Administration and scoring manual*. San Antonio, TX: Psychological Corporation.
- Wechsler, D. (2009). *Wechsler adult memory scale – Fourth edition*. San Antonio, TX: Pearson.
- Wiemers, E. A., Redick, T. S., & Morrison, A. B. (2019). The influence of individual differences in cognitive ability on working memory training gains. *Journal of Cognitive Enhancement*, 3, 174–185. doi:10.1007/s41465-018-0111-2
- Witlock, L. A., McLaughlin, A. C., & Allaire, J. C. (2012). Individual differences in response to cognitive training: Using a multi-model, attentionally demanding game-based intervention for older adults. *Computers in Human Behavior*, 28, 1091–1096. doi:10.1016/j.chb.2012.01.012
- Zinke, K., Zeintl, M., Rose, N. S., Putzmann, J., Pydde, A., & Kliegel, M. (2014). Working memory training and transfer in older adults: Effects of age, baseline performance, and training gains. *Developmental Psychology*, 50, 304–315. doi:10.1037/a0032982